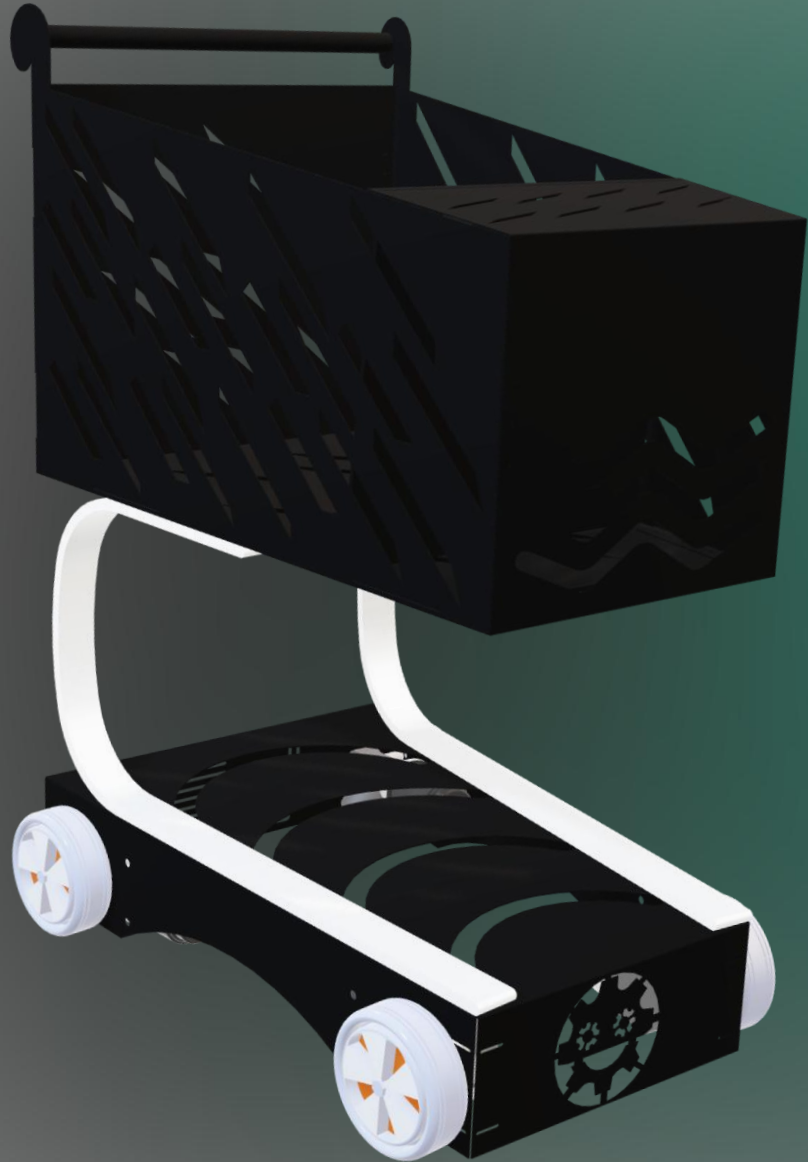




Project Overview

The **Automated Follow-Me Cart** combines **AI** and **computer vision** to assist elderly and disabled individuals with hands-free mobility. It autonomously follows users, avoids obstacles, and recognizes gestures. Paired with a **mobile app**, it offers features like battery monitoring, store navigation, item tracking, and remote control, ensuring convenience and full user control. Designed for various environments, the cart promotes independence and inclusivity through innovative technology.





Project Goal

Enhancing Independence and Convenience

To create a smart, hands-free mobility solution that increases independence, safety, and ease of use in daily activities.



The automated **Follow me** cart solution



Solution

An intelligent cart with automated motion, gesture-based control, and mobile app feedback to ensure a seamless and user-friendly experience.



User end Features

A Comprehensive system



Manual Control Section

Allows users to operate the cart manually when needed.



Autonomous Navigation Section

Utilizes a camera for automated motion, following the user seamlessly.



Feedback and Monitoring Section

Provides real-time monitoring and control via a mobile app.



Follow-Me Carty App!
Empowering your mobility
with AI and seamless
control

UI

User interface



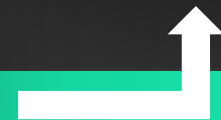
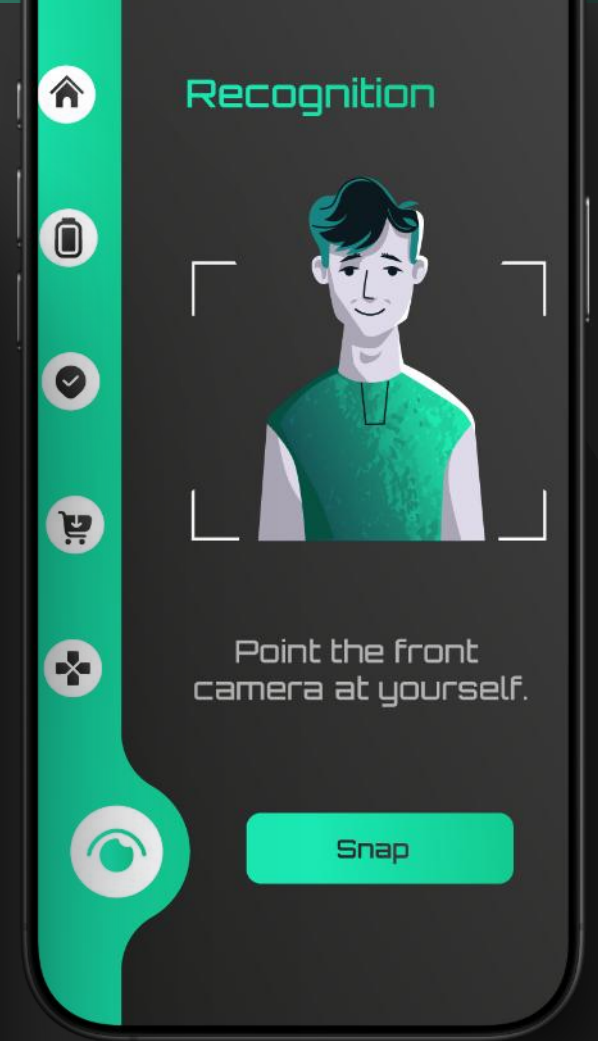


Works

1. Register
2. Recognition

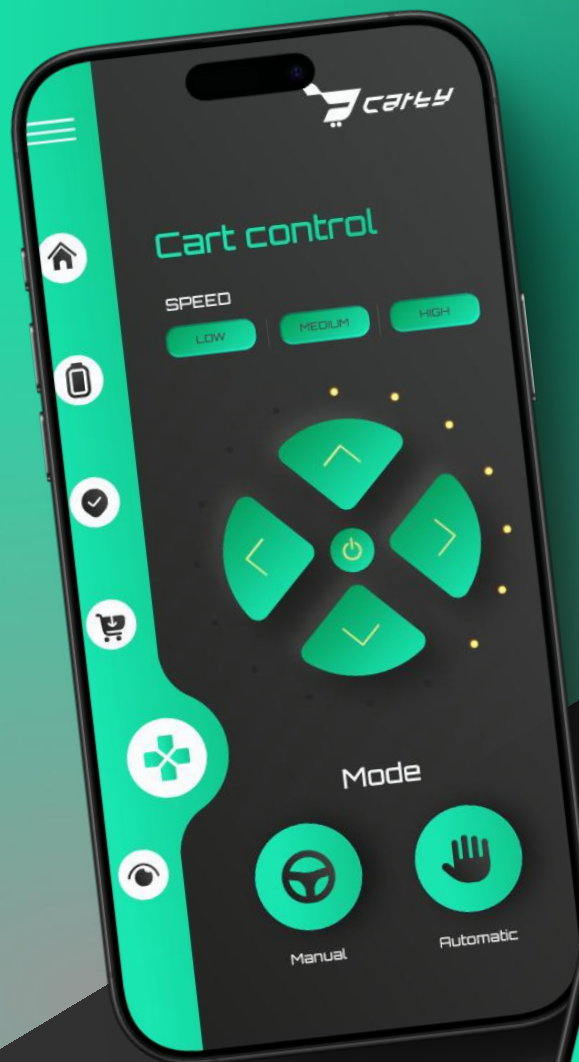
UI

User interface





Control



Navigation





Cart Content



Battery

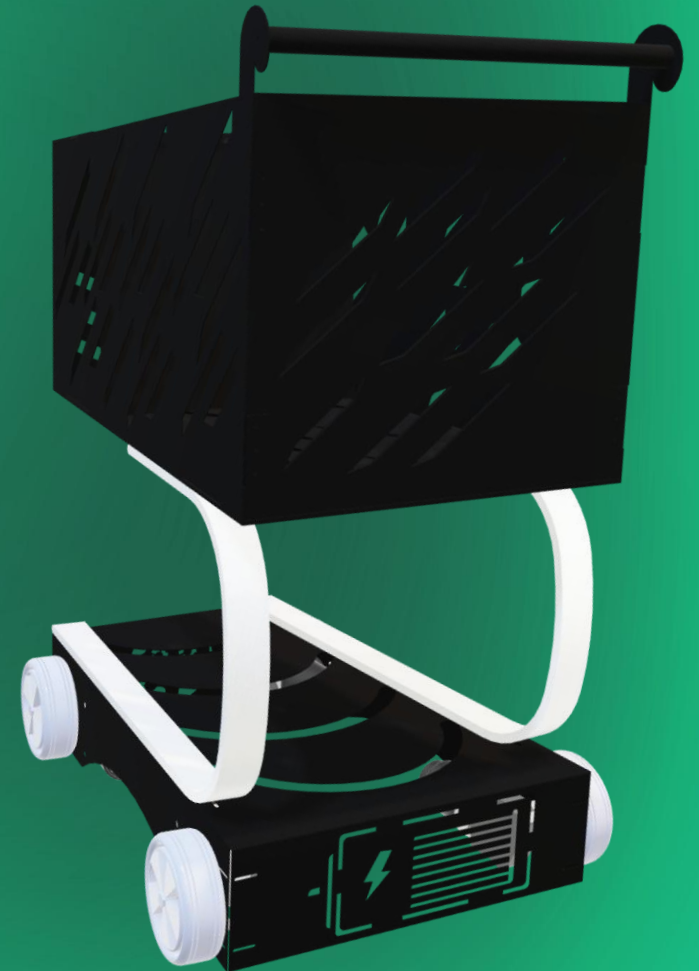
Key Sections

Of the Project

01

Mechanical Design

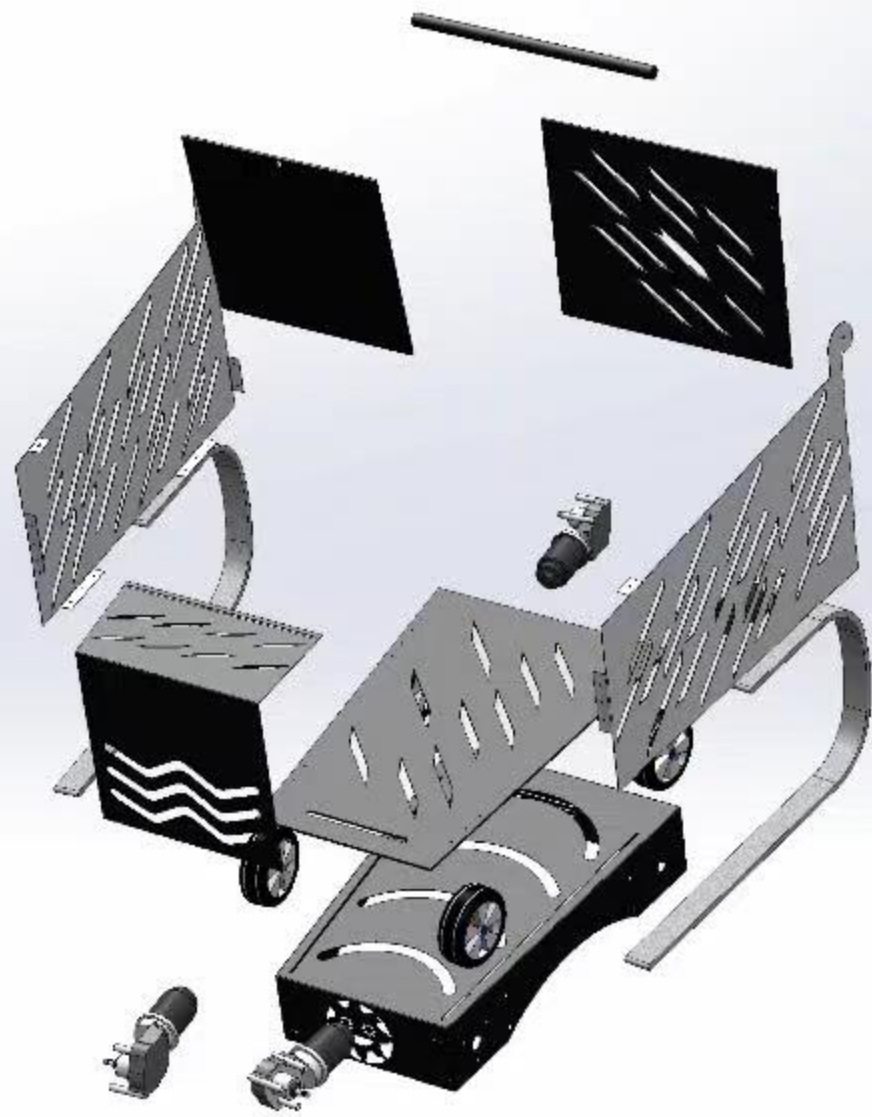
- **Development of a robust and portable structure using SolidWorks.**
- **Materials: Sheet metal for durability and wood for lightweight insulation.**
- **Integration of motors, wheels, and structural components for stability and maneuverability.**

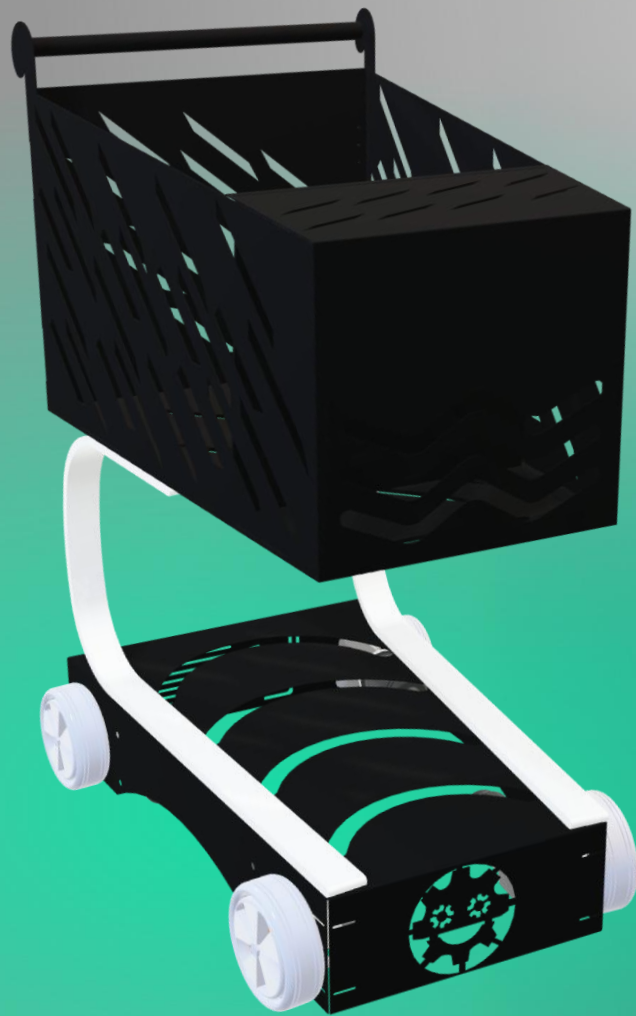




VS

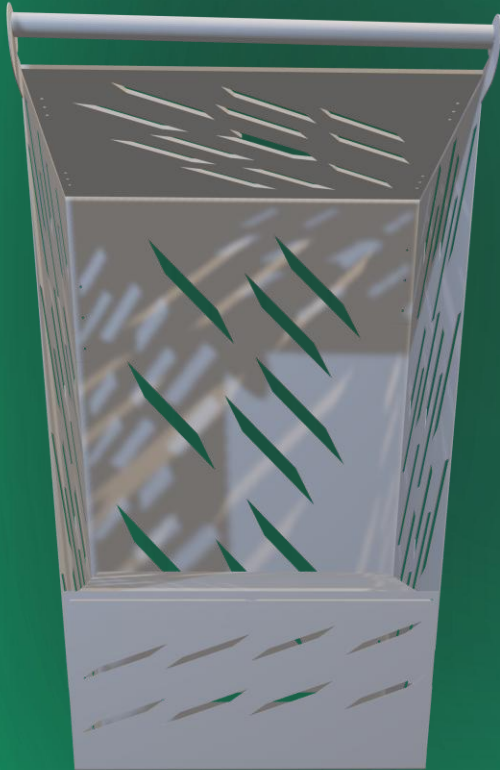




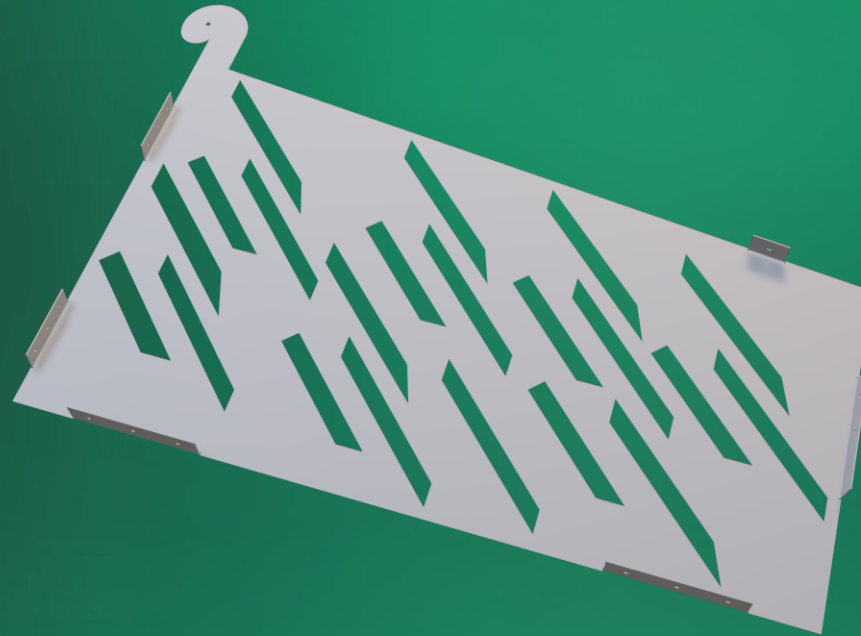


 **SOLIDWORKS**

1. Upper Body

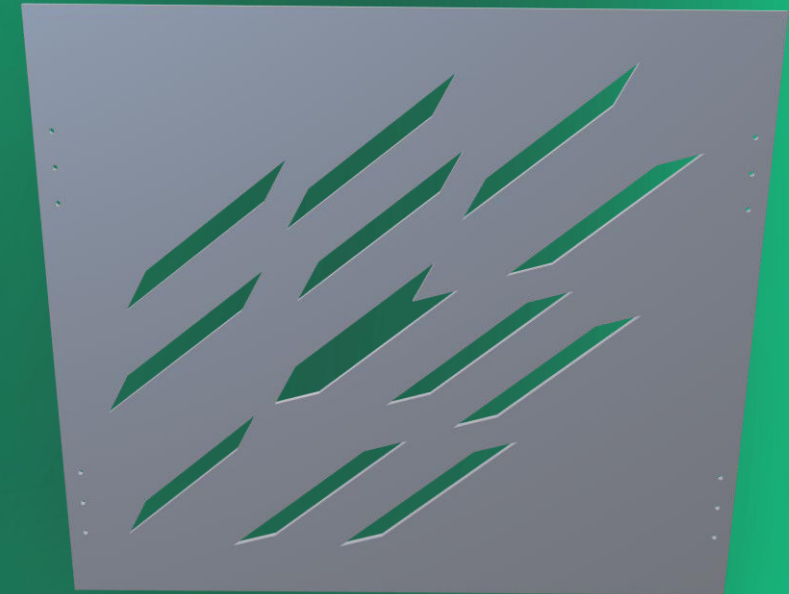


Full Upper Body (Top View)



Right & Left Side Upper

Material → Sheetmetal 1mm
Angles design 87.5°



Back Upper

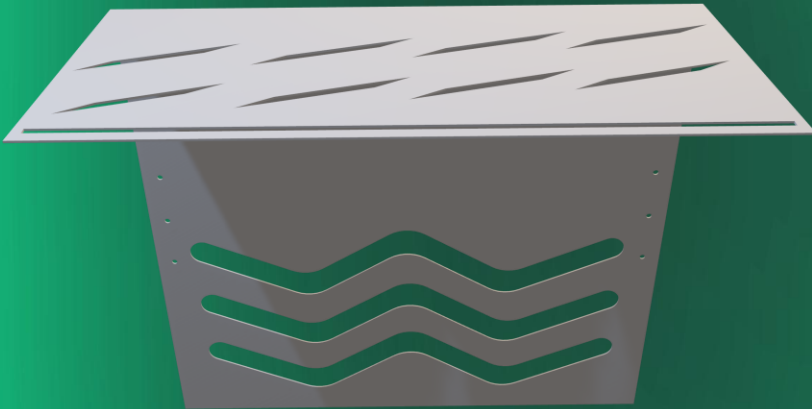
Material → Laser Cut Wood 4mm

1. Upper Body



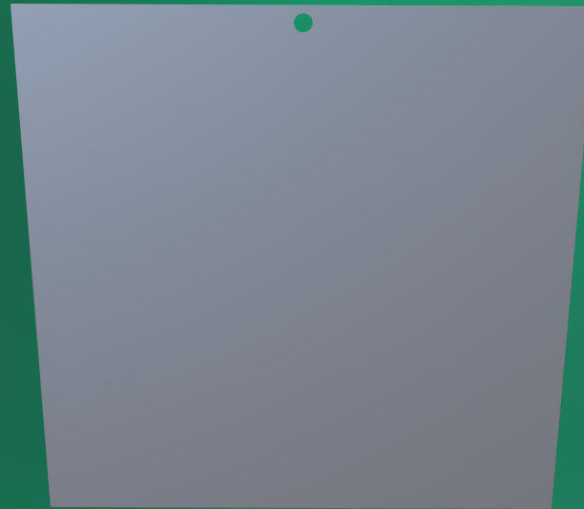
Holder

Material → Wood 1.5cm
Diameter



Face Upper

Material → Sheetmetal
1mm



Closing Sheet for
Hardware
Components

Material → Laser Cut Wood
4mm



Full Upper Body (Side View)

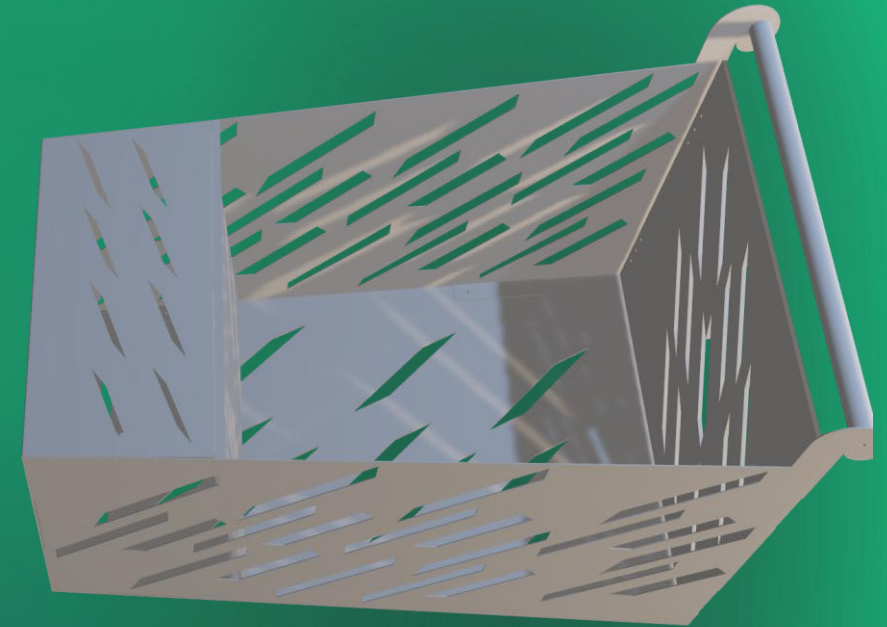
1. Upper Body

Upper Body Design Results in:

- Area for Products
 $45 \times 40 \text{ cm}^2$
- Area for Hardware Components
 $15 \times 25 \text{ cm}^2$

Extra in Design:

- Multi Layers for Hardware Components → Threaded Rods
- Camera Place

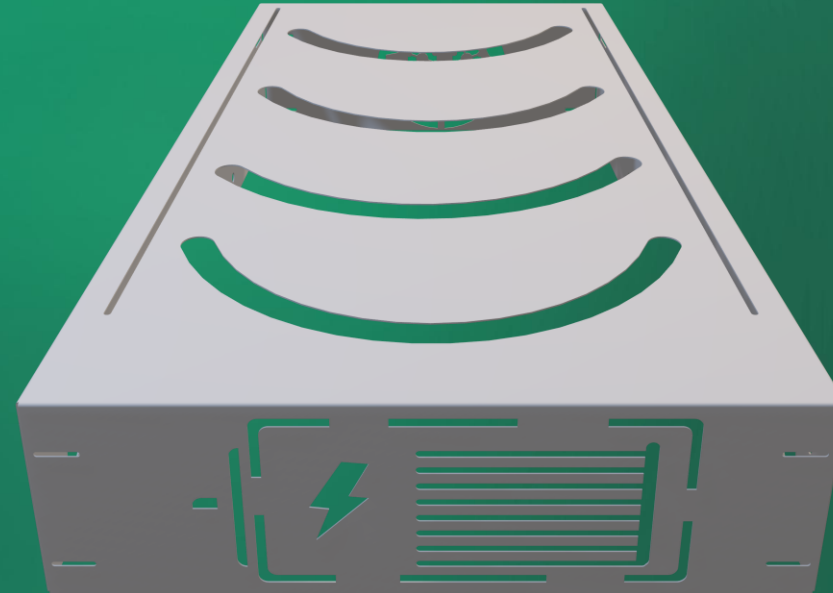


Full Upper Body (Isometric View)

2. Lower Body

Why Galvanized Iron?

- Reliable
- Can sustain heavy weights
- Tough



Lower Body (Isometric Top View)

Material → Sheetmetal (Galvanized Iron) 1.5mm

3. Linking Part & Threaded Rods

Material → Elastic Metal

Threaded Rods → Stability
→ Wires Way



Threaded Rods



Linking Part (Isometric View)

4. Wheels & Couplers



Wheel (Isometric View)



Coupler Layout

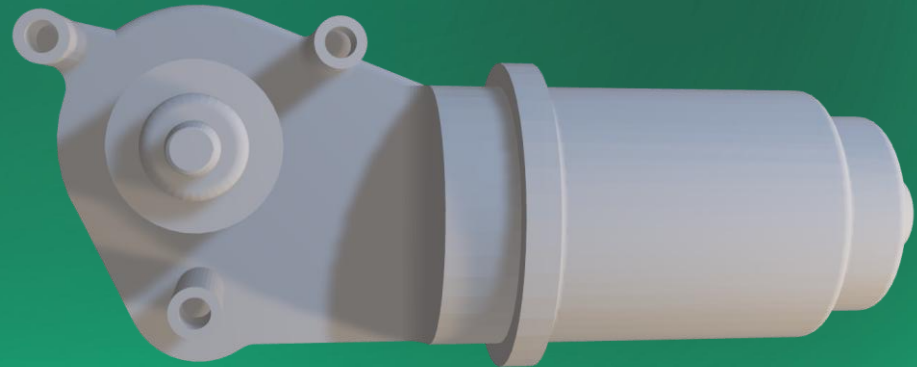
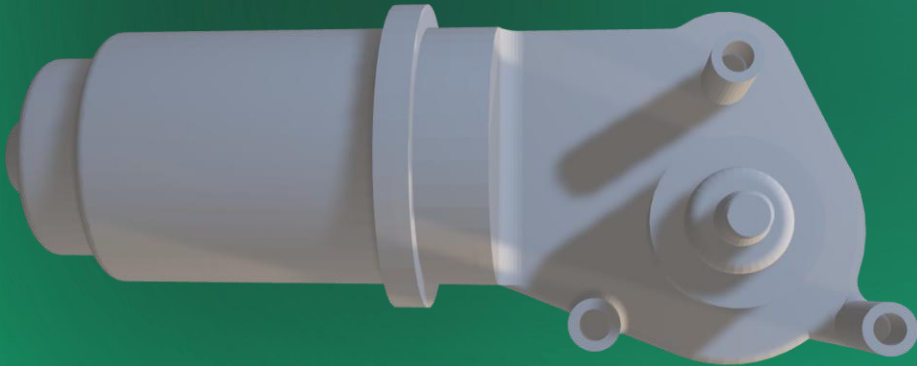
5. Motors

4 Wiper Motors

- For smaller passenger vehicles
Torque = 10 : 20 Nm
- For larger vehicles (SUVs, trucks, or buses)
Torque = 30 : 50 Nm or higher



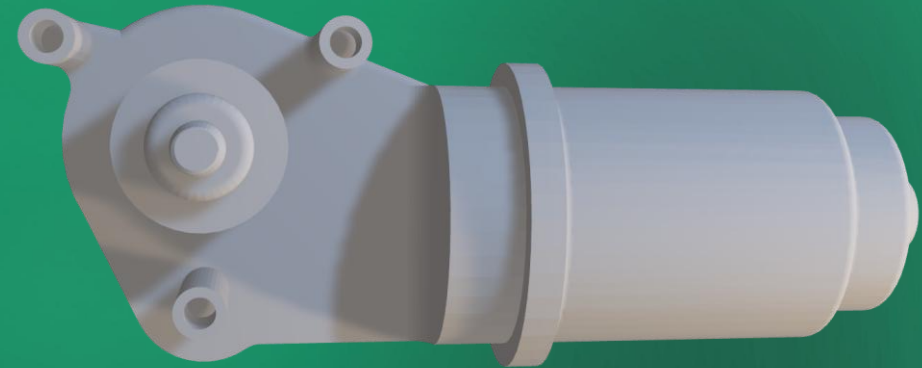
Wiper Motor Layout



5. Motors

Given Data:

1. Torque per motor (TTT) = 20 Nm
2. Number of motors = 4
3. Wheel diameter (d) = 13 cm = 0.13 m
4. Gravitational acceleration (g) = 9.81 m/s²



- **Step 1:**

Total Torque The total torque provided by all motors is:

$$T_{\text{total}} = 4 \times 20 = \mathbf{80 \text{ Nm}}$$

- **Step 2:**

Force at the Wheel The force exerted at the edge of the wheel is: Since,

$$F = T_{\text{total}} / r \text{ Then, } F = 80 / 0.13 \approx \mathbf{615.38 \text{ N}}$$

- **Step 3:**

Maximum Weight Convert the total force into mass:

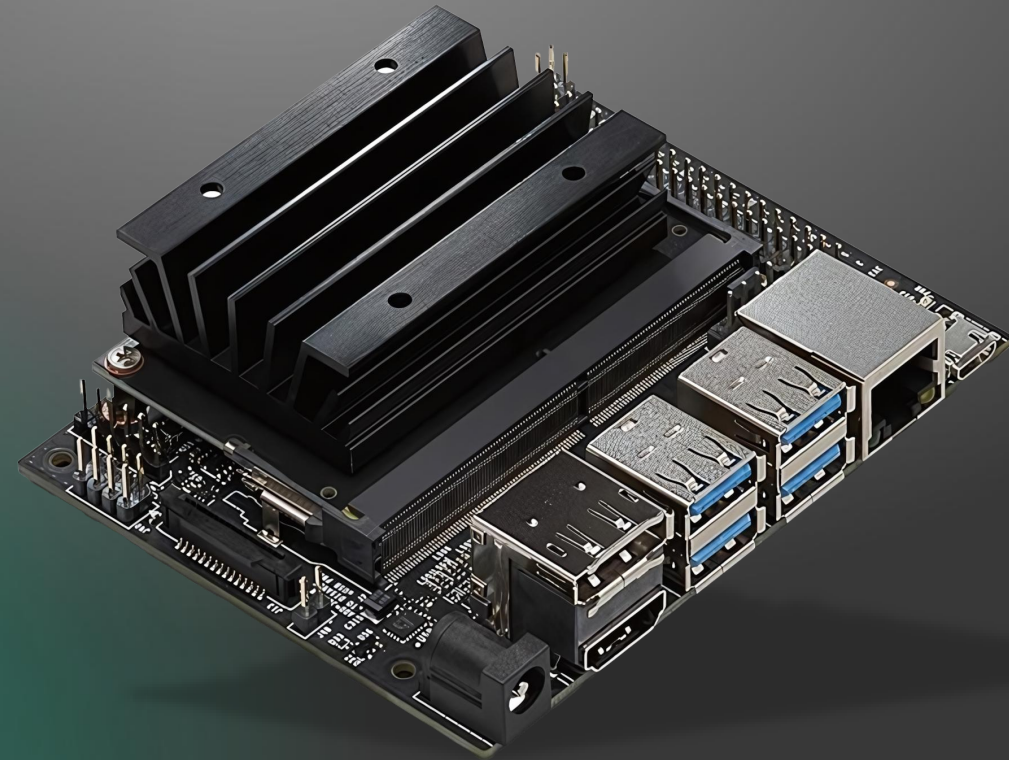
$$m = F / g \text{ then,}$$

$$F = 615.28 / 9.81 = \mathbf{62.7 \text{ kg}}$$

02

Hardware Implementation

- **Core Components:** Motors, IR & ultrasonic sensors, camera, Arduino Nano, and NVIDIA Jetson Nano.
- **Power System:** Lithium-ion battery with Battery Management System (BMS).
- **Efficient wiring and modular PCBs** for seamless operation.

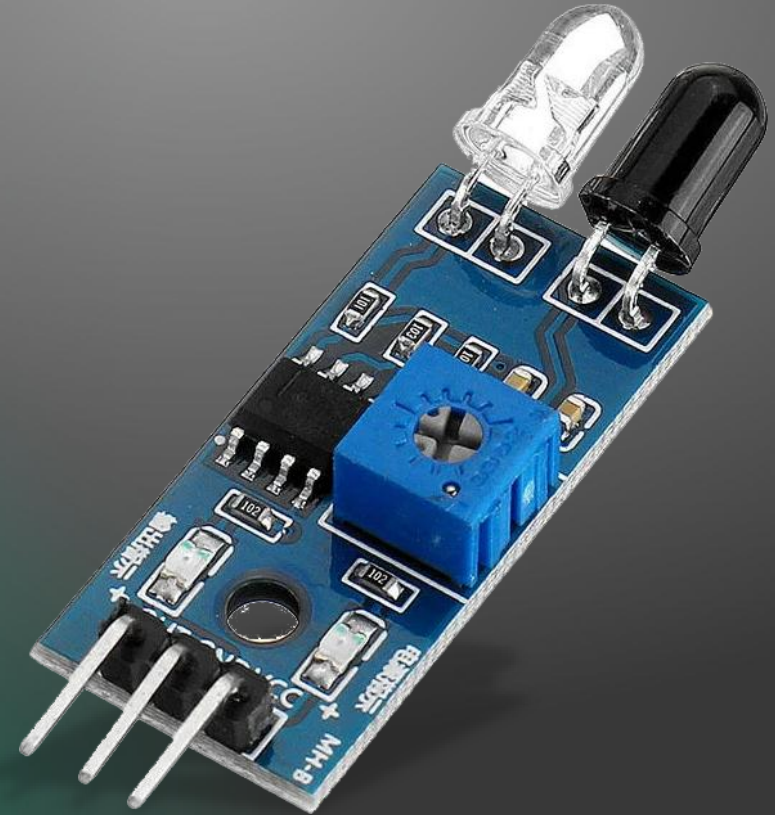


Core Components

Ir Sesnsor

- Specifications:
- Detection Range: 10 cm to 20 cm
- Operating Voltage: 5V.
- Response Time: <1 ms.

Placement front and sides



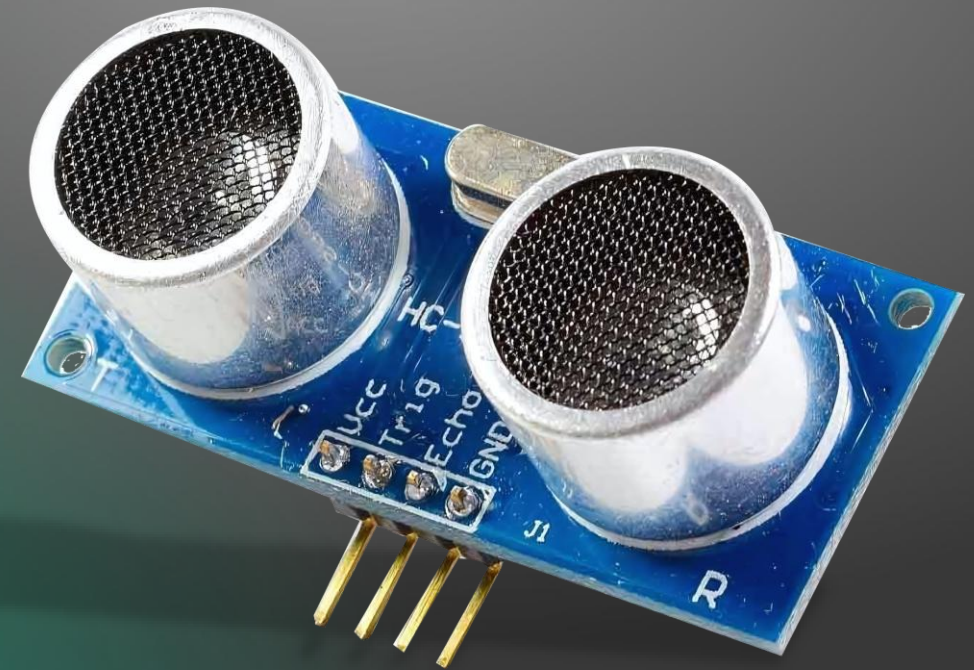
Core Components

Ultrasonic Sesnsor

Specifications:

- Detection Range: 2 cm to 4 m.
- Operating Voltage: 5V.
- Accuracy: ± 3 mm..

Placement front



Core Components

Joystick module

Specifications:

- **Operating Voltage: 5V.**

Outputs:

- **X-axis (VRx):** Analog output (horizontal movement).
- **Y-axis (VRy):** Analog output (vertical movement).
- **Button (SW):** Digital output (press detection).
- **Pins:** GND, +5V, VRx, VRy, SW for easy connectivity.



Core Components

Arduino Nano

Specifications:

- Resolution: 1920 x 1080 pixels.
- Frame Rate: 30 fps.
- Field of View: 120 degrees.

Functionality

Captures real-time video for gesture recognition and user tracking.



Core Components

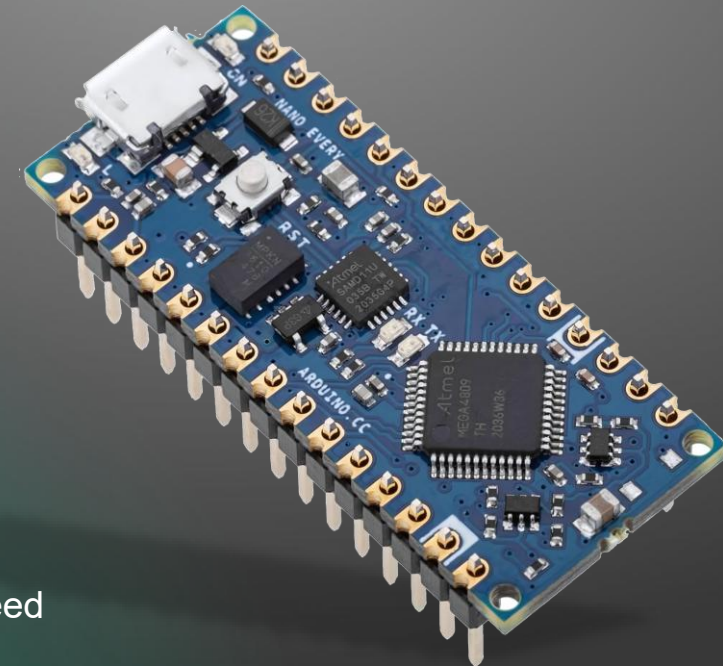
Arduino Nano

Specifications:

- Processor: ATmega328P.
- Operating Voltage: 5V.
- Digital I/O Pins: 14.

Functionality

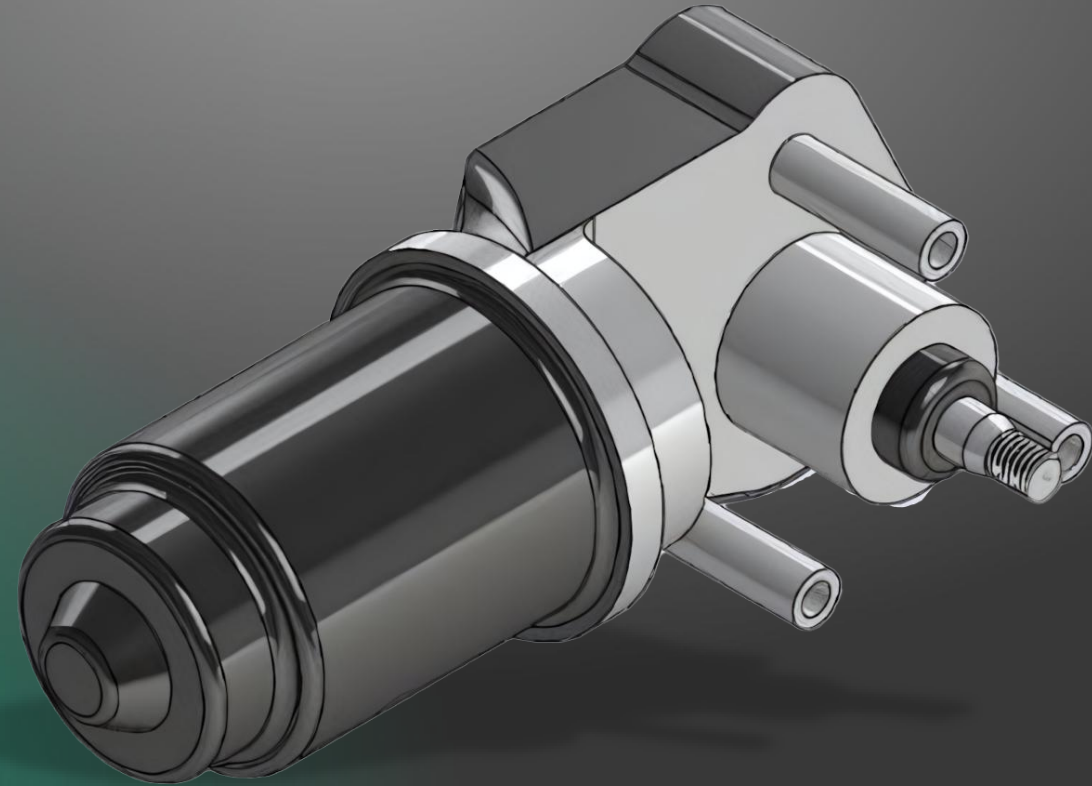
The Arduino Nano manages auxiliary functions, such as motor speed adjustments and sensor data acquisition.



Core Components

Wiper motors

- Rated Torque: 20 Nm per motor.
- Maximum Operating Voltage: **12V**.
- Operating Current: 4-5A underload.
- Motor Efficiency: Approximately 85%.
- Maximum Power consumption 192 watt



Power System

Specifications:

- Type: Lithium-ion.
- Model: Royal Battery 12n9-bs.
- Capacity: 12V, 32Ah.

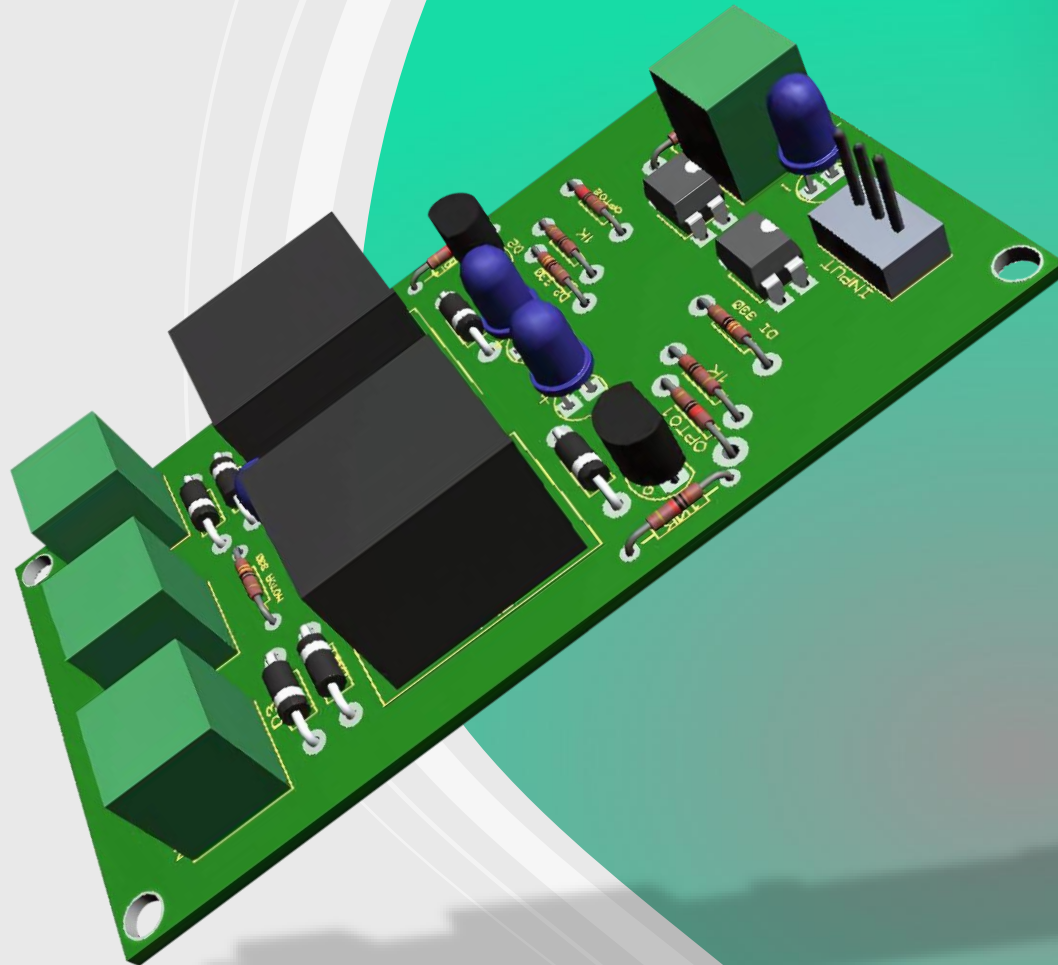


PCB Design
APPs



PROTEUS

Motor driver PCB

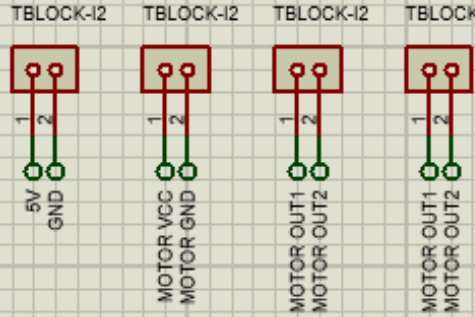


01

Functionality

Direction Control: .1

Digital Signals from Nano .2



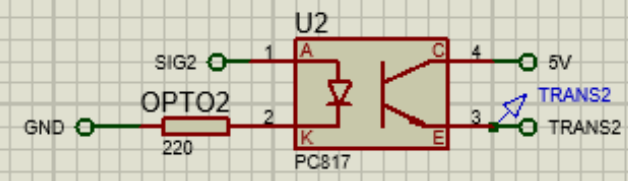
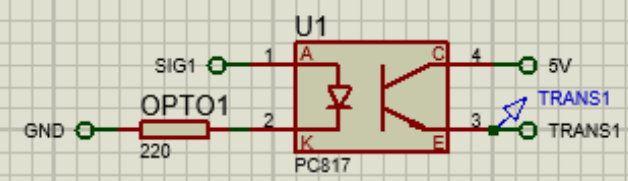
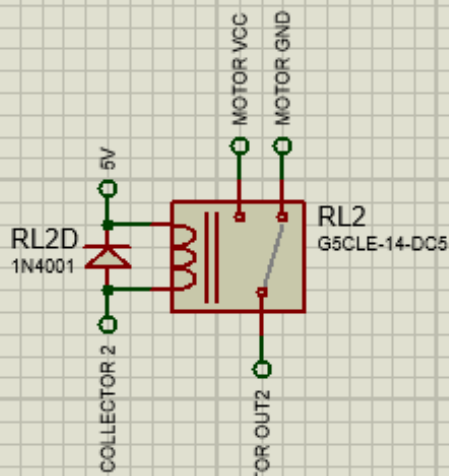
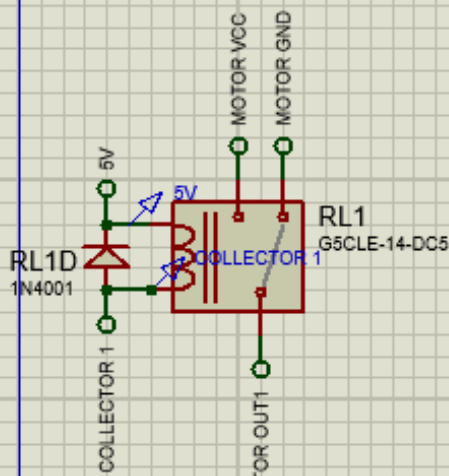
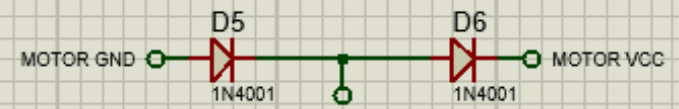
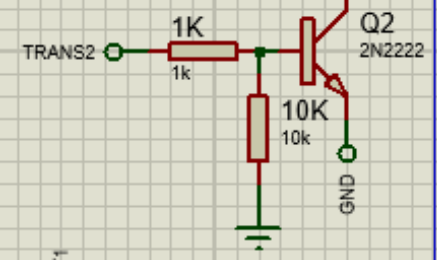
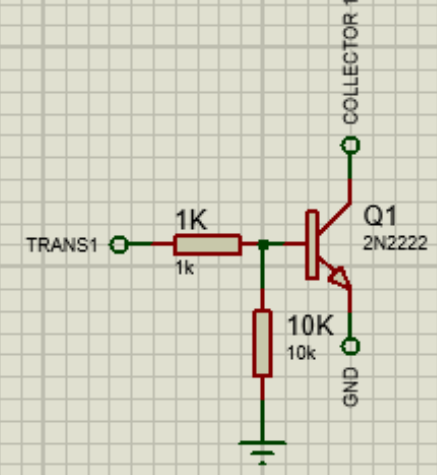
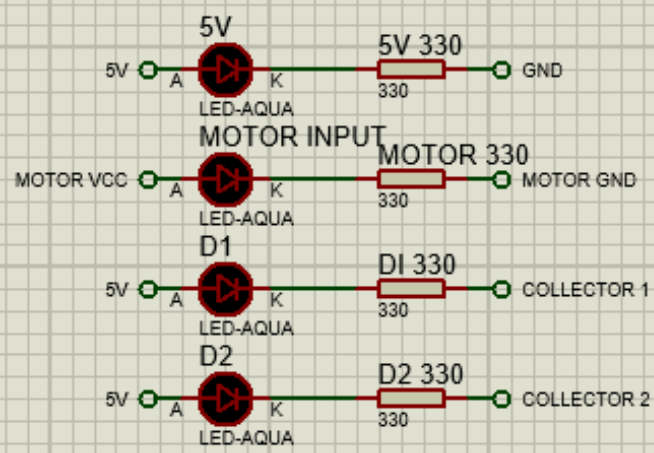
INPUT
SIL-100-03



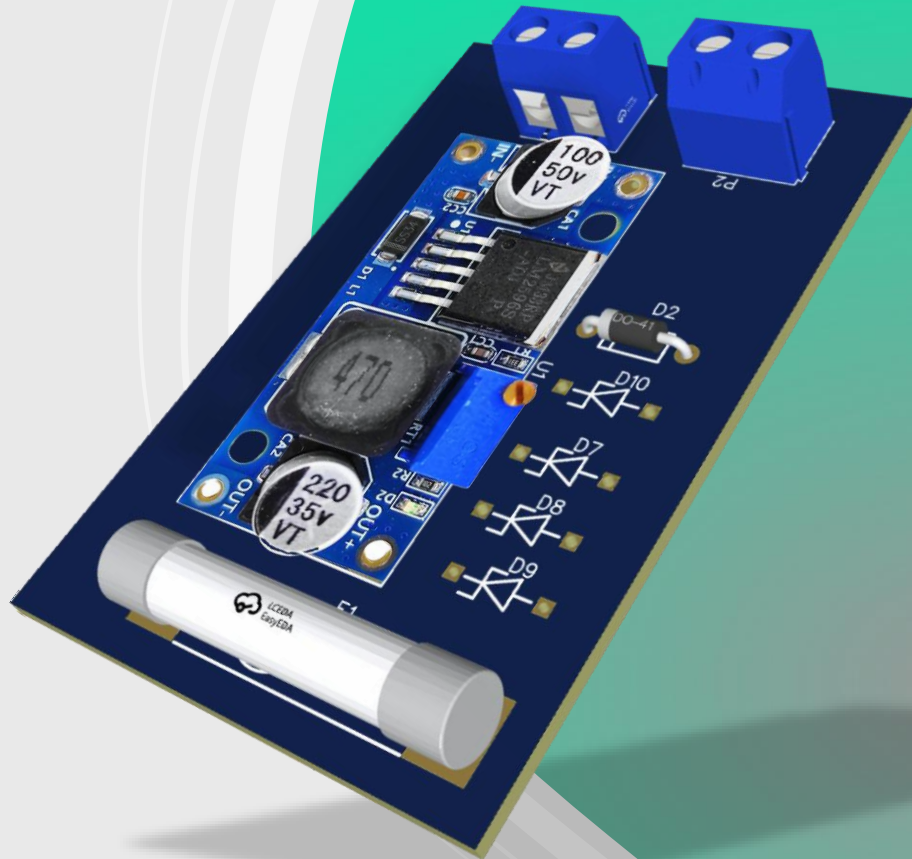
B1
5V



B2
12V



Safety PCB

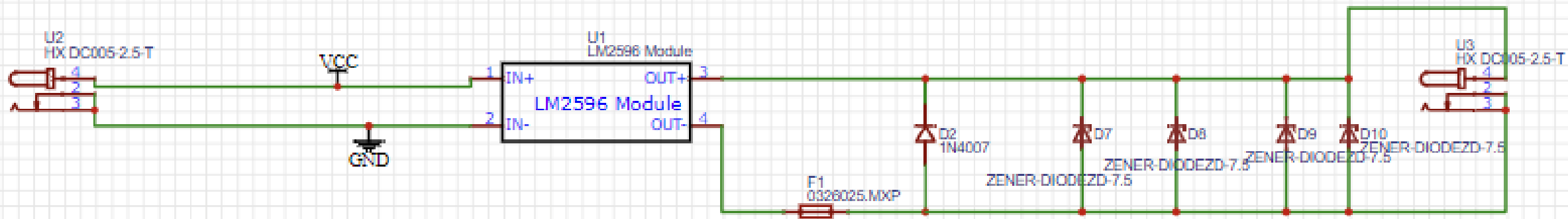


02

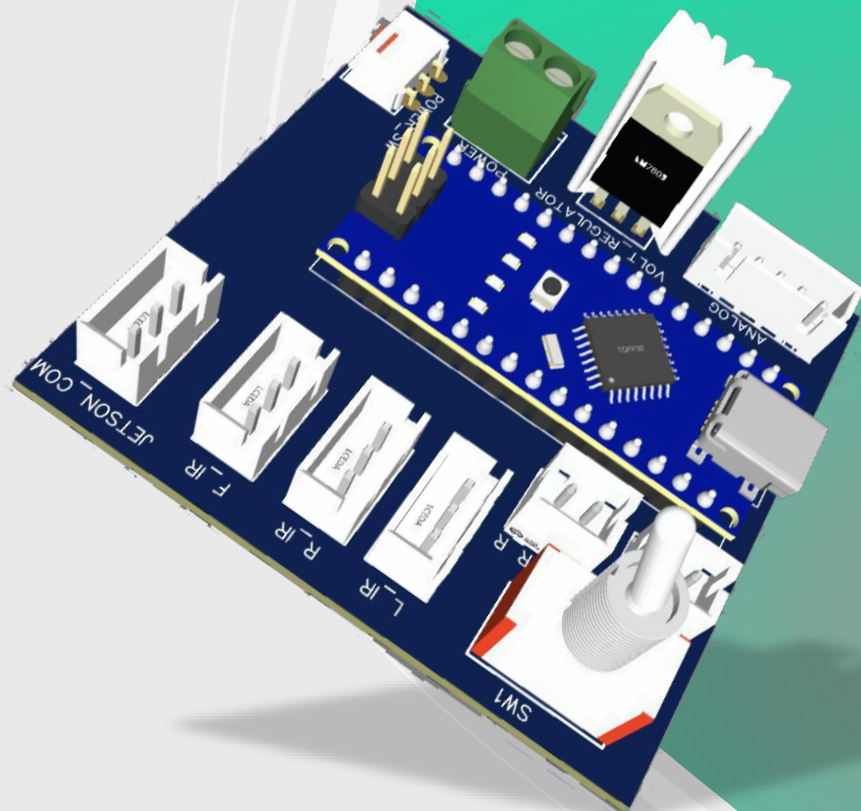
Functionality

Protection Mechanisms .1

Automated Disconnection .2



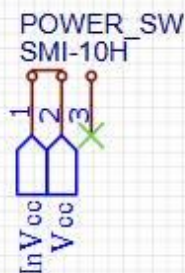
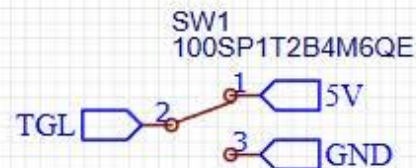
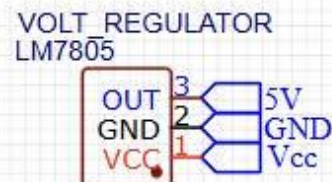
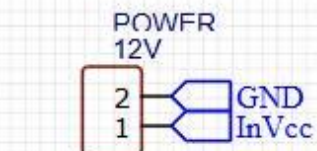
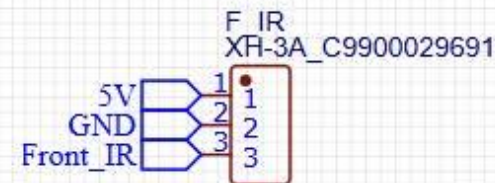
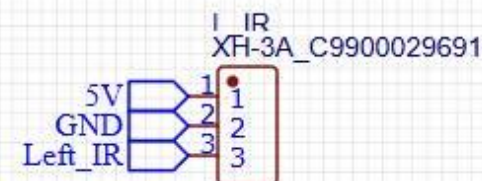
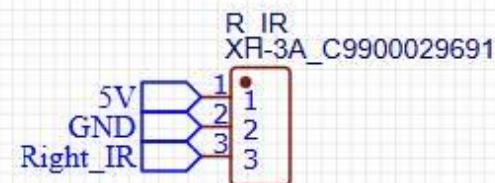
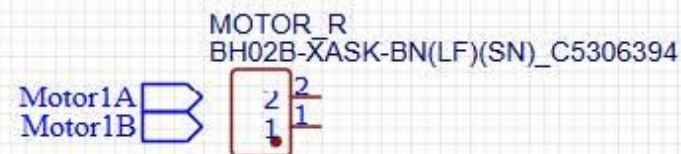
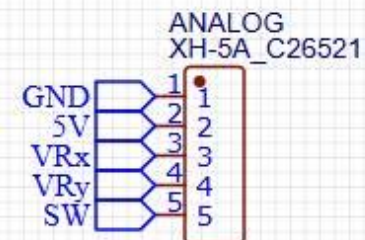
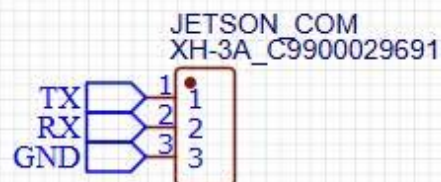
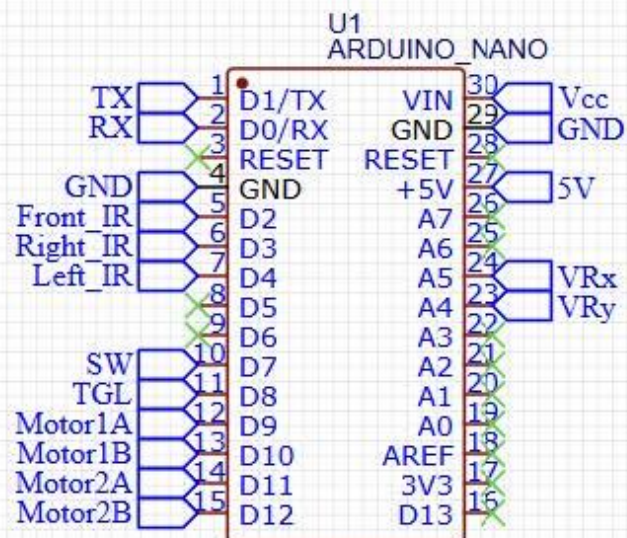
Nano & Joystick PCB



03

Functionality

Joystick Input	.1
Motor Control	.2
Sensor Integration	.3
Jetson Nano Communication	.4
Power Management	.5
Real-Time Coordination	.6



03

Software Development

Requirements:

Real-time performance on embedded devices. High accuracy for gesture recognition and clothing detection under various conditions. Low computational overhead for smooth operation on limited hardware.

Development Process:

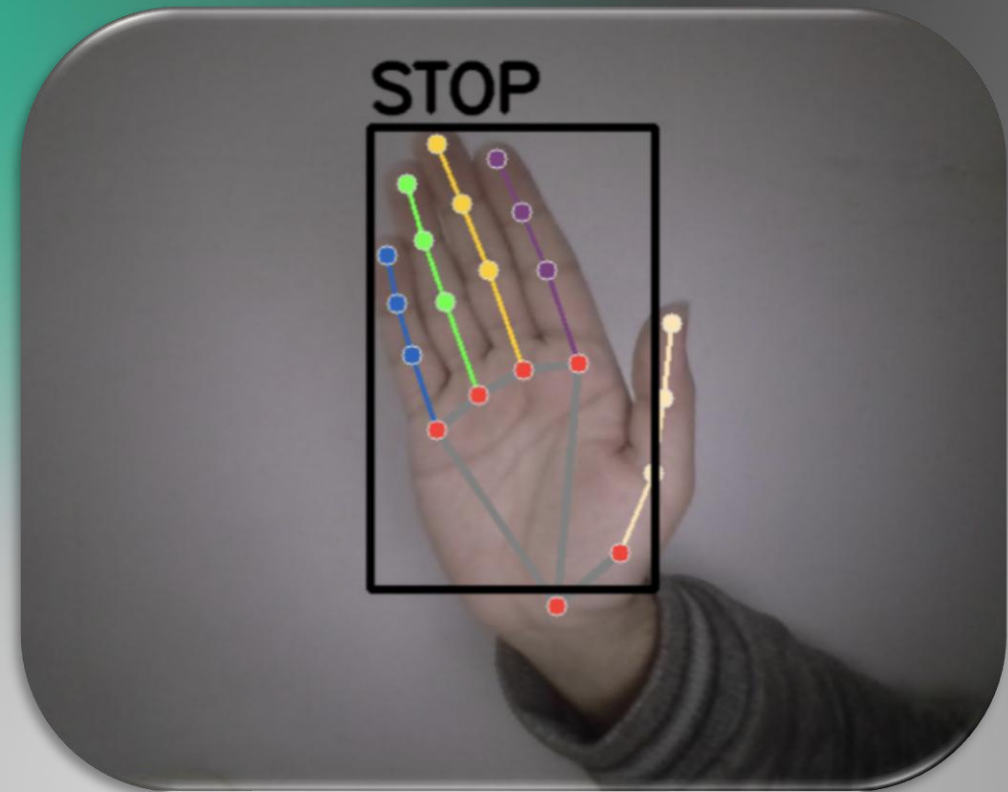
Followed the V-Model for development



Development of Key Models



A. Hand Gesture Recognition





Why Mediapipe?

Lightweight and highly optimized for embedded devices. Provides accurate tracking of 21 hand landmarks, simplifying gesture classification.

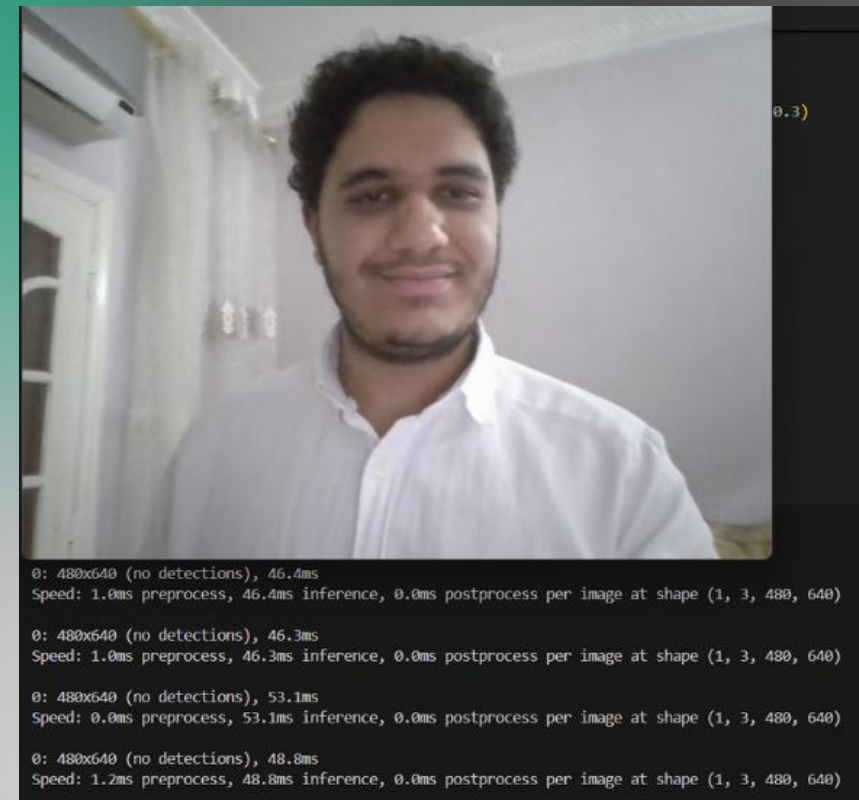
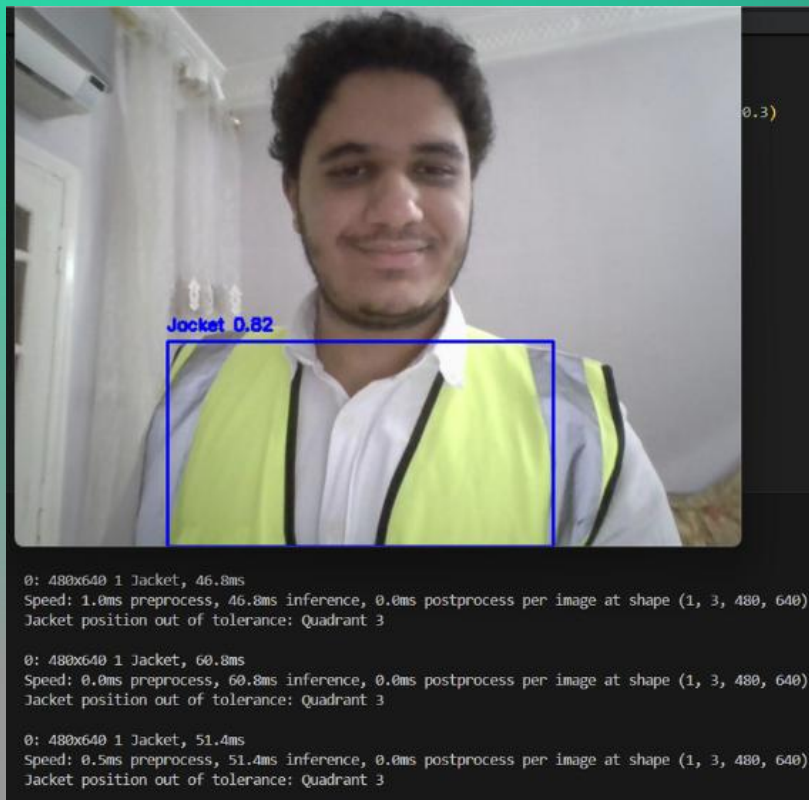
Development Process

Data Collection: Captured gesture data in diverse lighting and environmental settings. Gesture Mapping: Defined specific gestures (e.g., open hand, fist, pointing) and mapped them to cart commands. Testing: Evaluated system performance for gesture recognition under real-world conditions. Optimization: Tuned the system for real-time responsiveness and low power consumption.

Challenges

Hand occlusion: Addressed using Mediapipe's pre-trained models with advanced tracking logic. Environmental variability: Included diverse datasets to improve robustness.

Vest Detection Using yolov8





Why yolov8 ?

Combines speed and accuracy, critical for embedded systems. Efficient inference on edge devices with optimized architecture.

Development Process

Dataset Preparation:

Collected images of vests across different styles, colors, and perspectives. Annotated data for precise bounding box labeling.

Model Training:

Fine-tuned a pre-trained YOLOv8 model using the annotated dataset. Applied techniques like data augmentation to improve generalization.

Testing

Validated performance with real-world video streams.

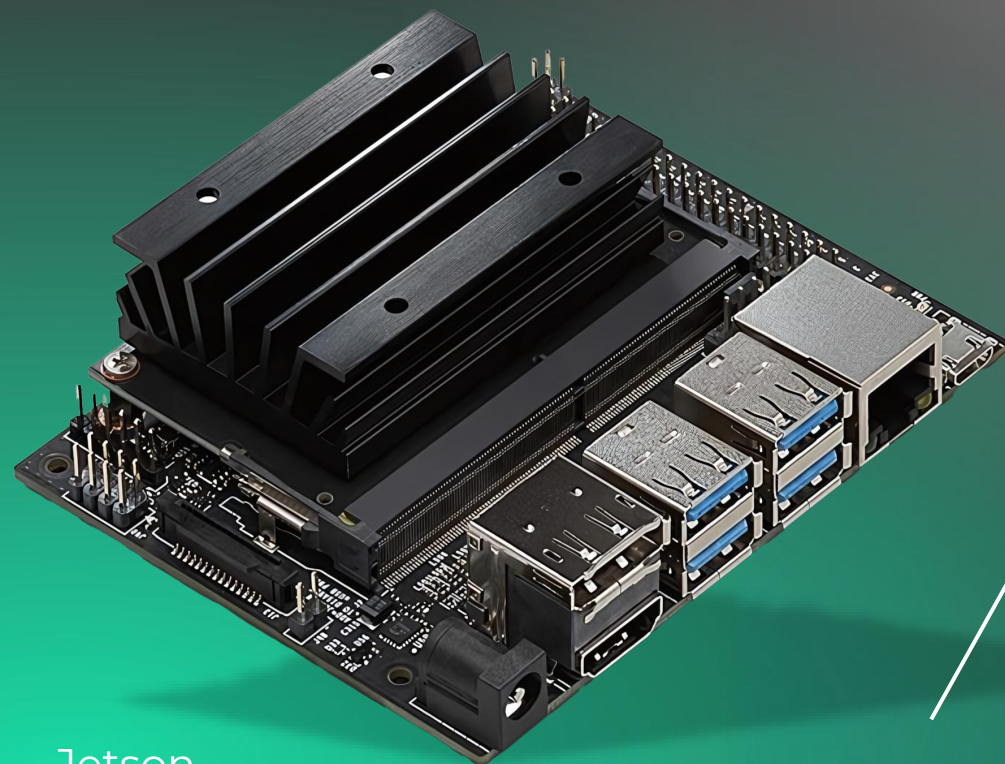
Computer vision models test



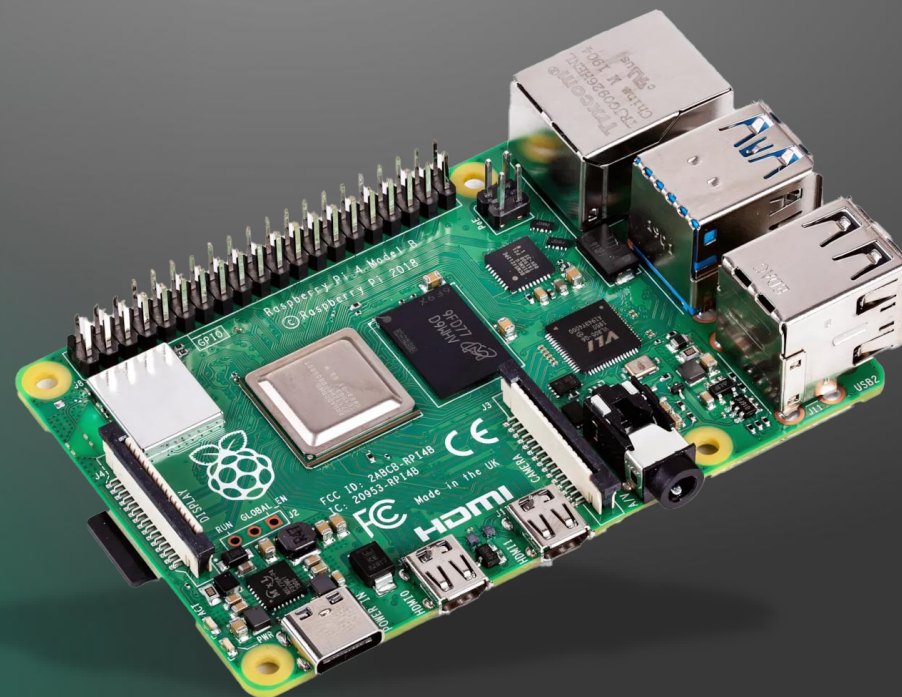
Why we used
NVIDIA Jetson

?

VS



Jetson



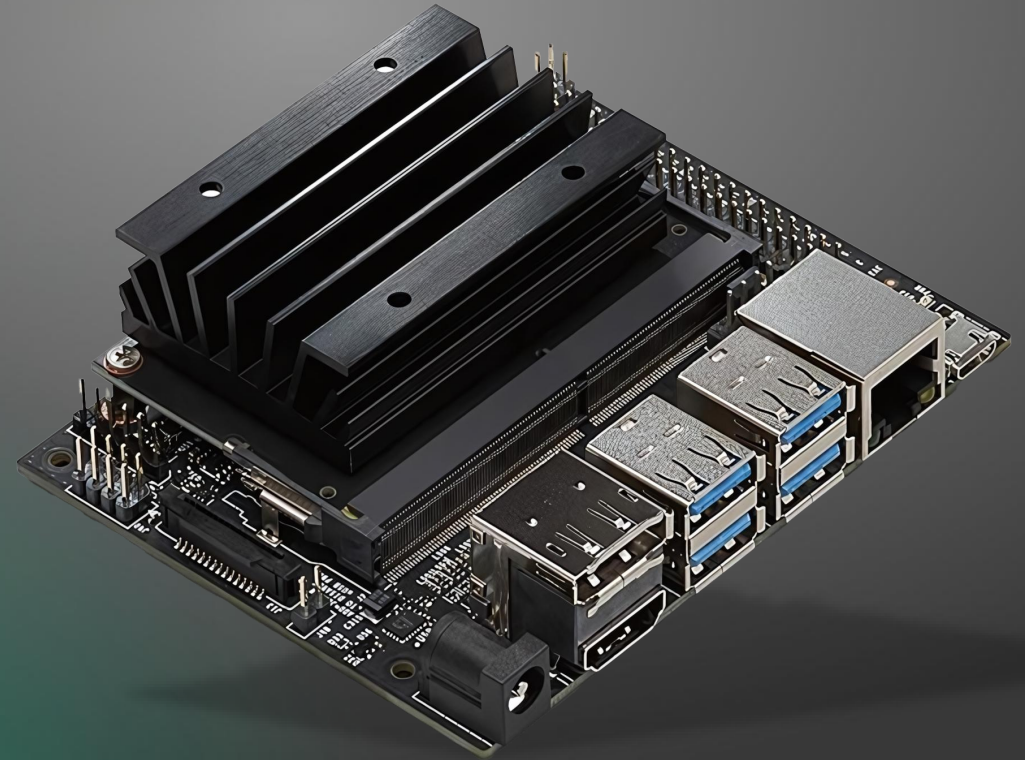
RaspberryPi

Core Components

Nvidia Jetson Platform

Specifications:

- GPU: 128-core Maxwell architecture.
- Memory: 4GB LPDDR4.
- Interfaces: CSI camera, USB 3.0, GPIO.



How to

Deal with

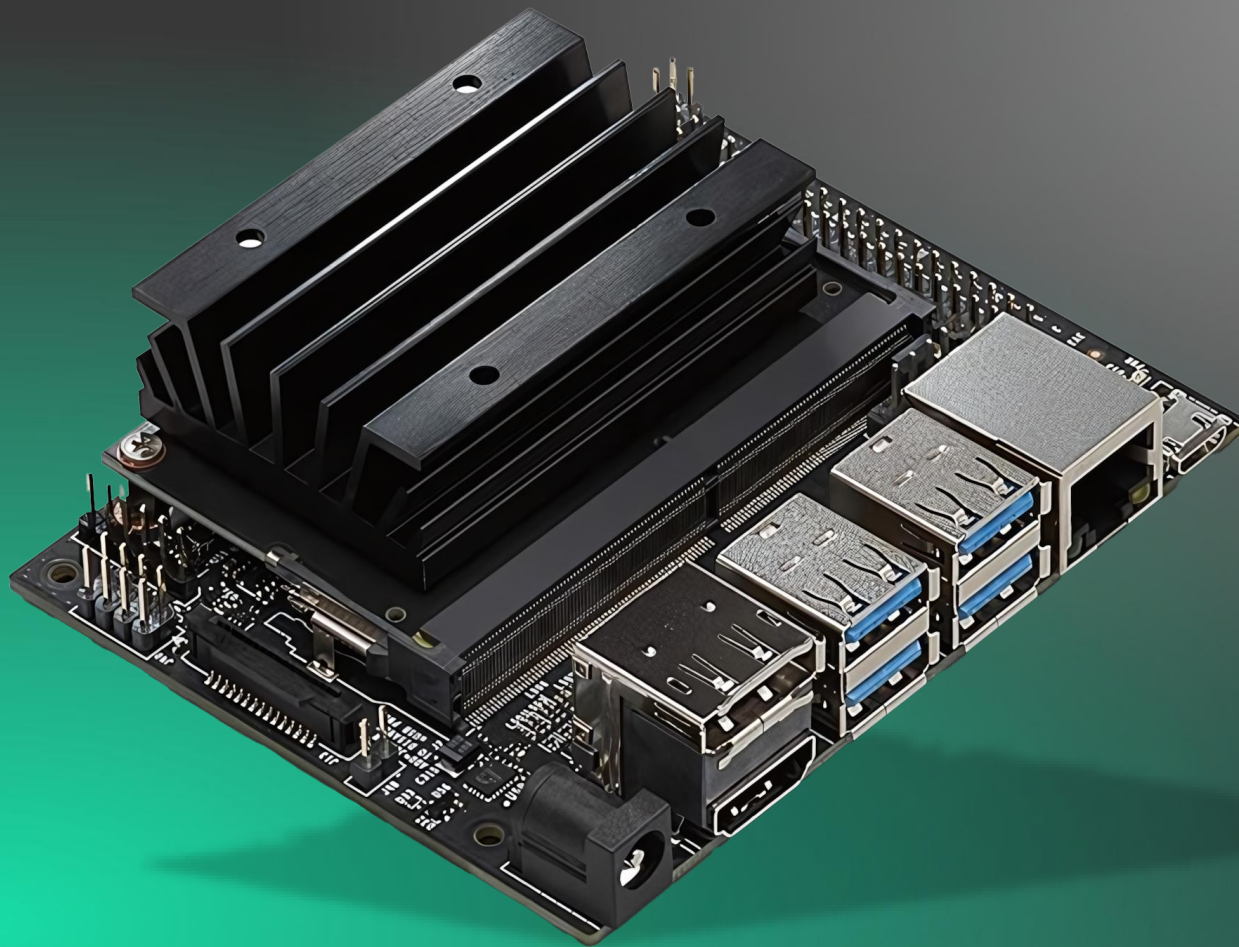
Jetson Nano

Jetson Setup

Jetpack •

SDK Image •

Operating system •



Initialization environment



Initialization environment





ANACONDA®



NumPy
1.23.5



8.3.15



PyTorch
2.5

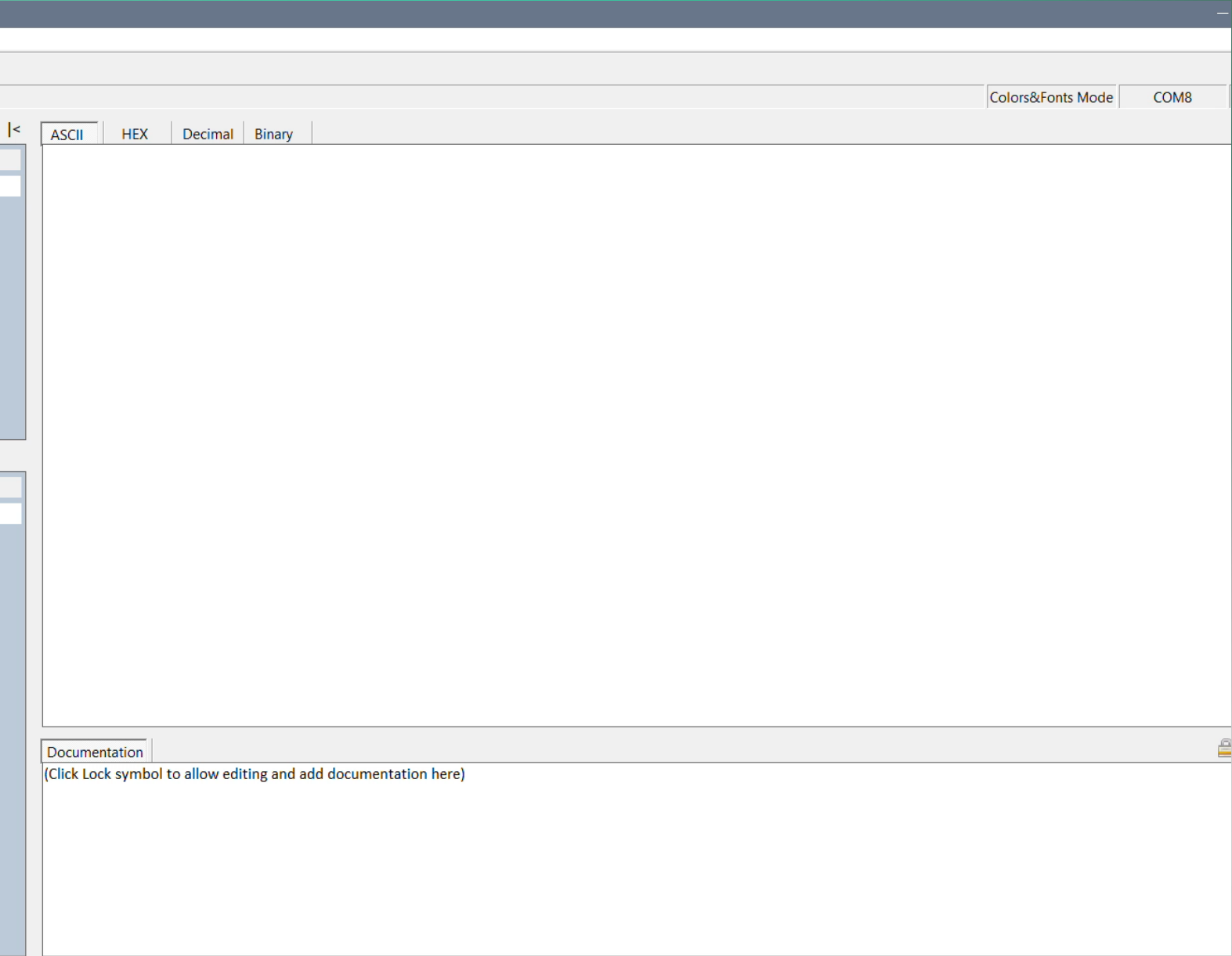


OpenCV
4.10.0

Pandas
2.2.3



Camara test:



ATmega328P Project Design Overview

1

Application Layer

Coordinates overall system behavior.

2

HAL Layer

Integrates peripherals (joystick, motors, sensors).

3

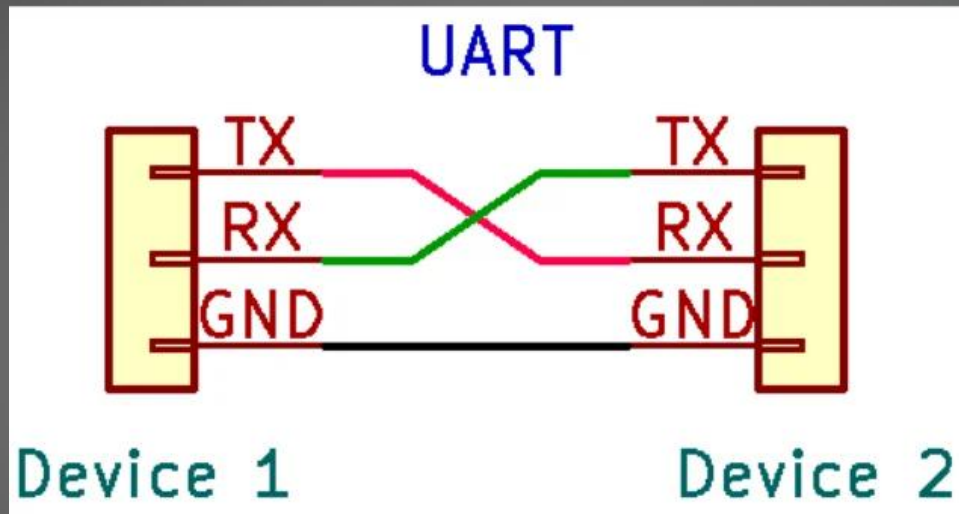
MCAL Layer

Handles low-level hardware interaction.

Mcal layer – uart Initialization

Why UART?

Simple protocol ideal for short distances and low data rates.



Configuration Steps

1. Baud Rate: 9600 bps (sufficient for project requirements).
2. Data Bits: 8 bits for effective communication.
3. Parity Bit: None (simplifies the protocol).
4. Stop Bits: 1 stop bit (signals end of each frame).

Result: UART configuration ensures reliable communication for the project.

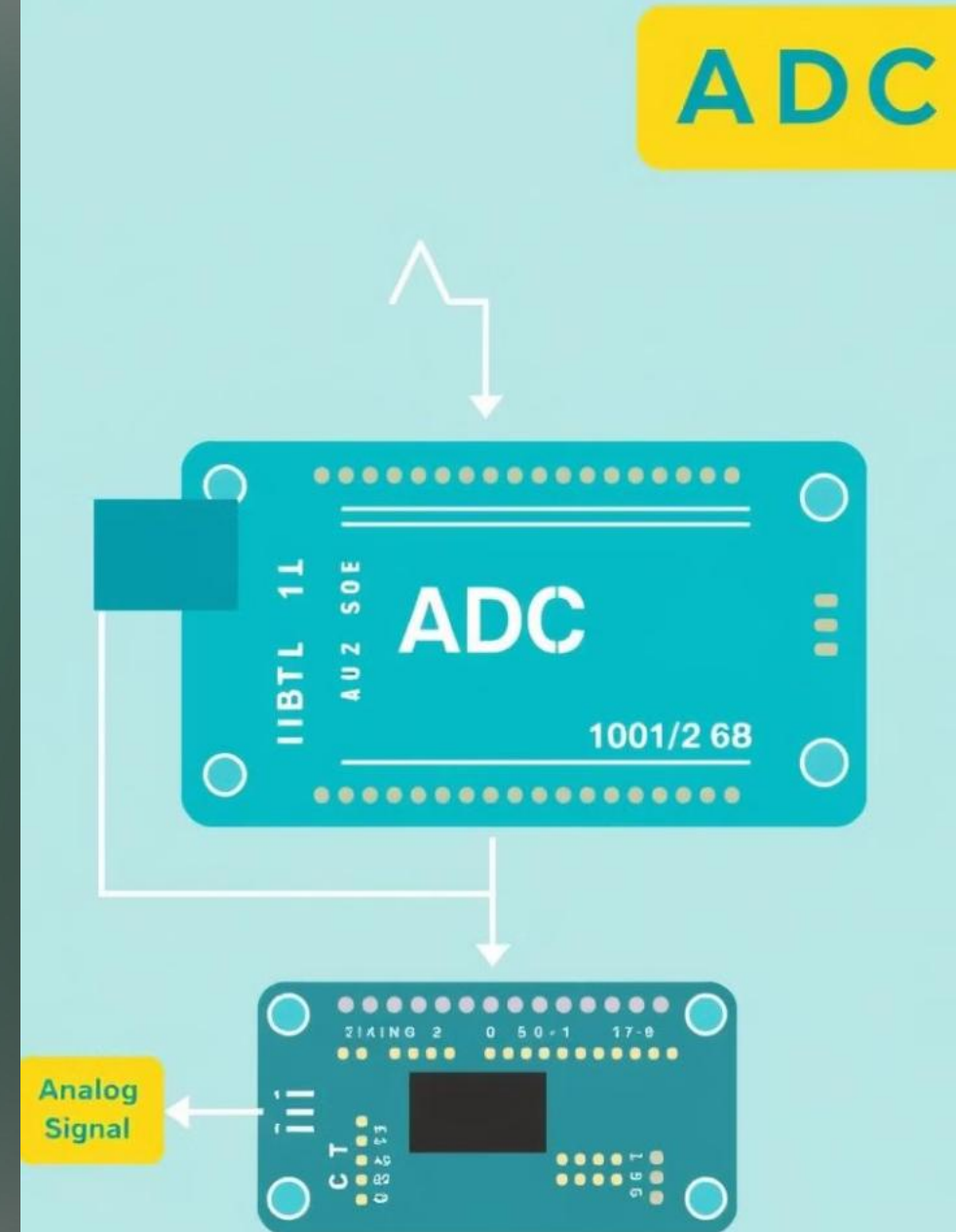
Mcal Layer-adc Configuration

Purpose

Converts analog signals from sensors and joystick into digital values for processing.

key benefits

Enables accurate interpretation of user inputs and sensor data.
Ensures seamless integration with the microcontroller for further processing.



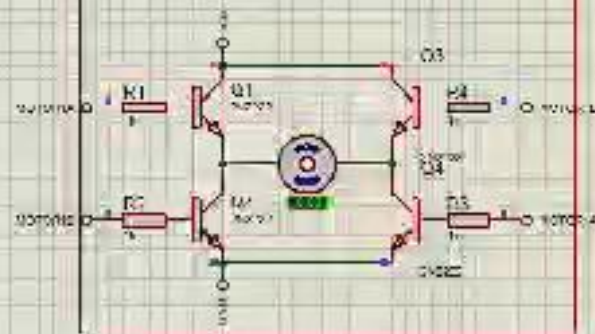
Proteus simulation test:

Analog module



/* S F B R L H G N */

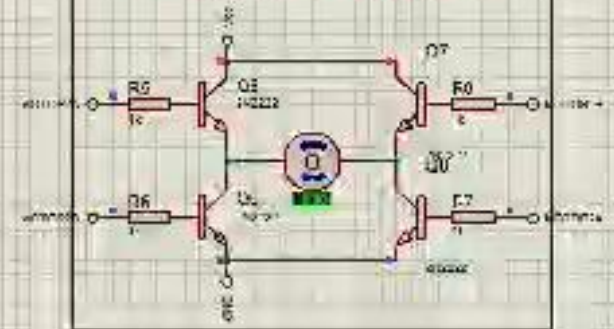
Left Motors



Controller



Right Motors



hal & application Layers

1

HAL Layer

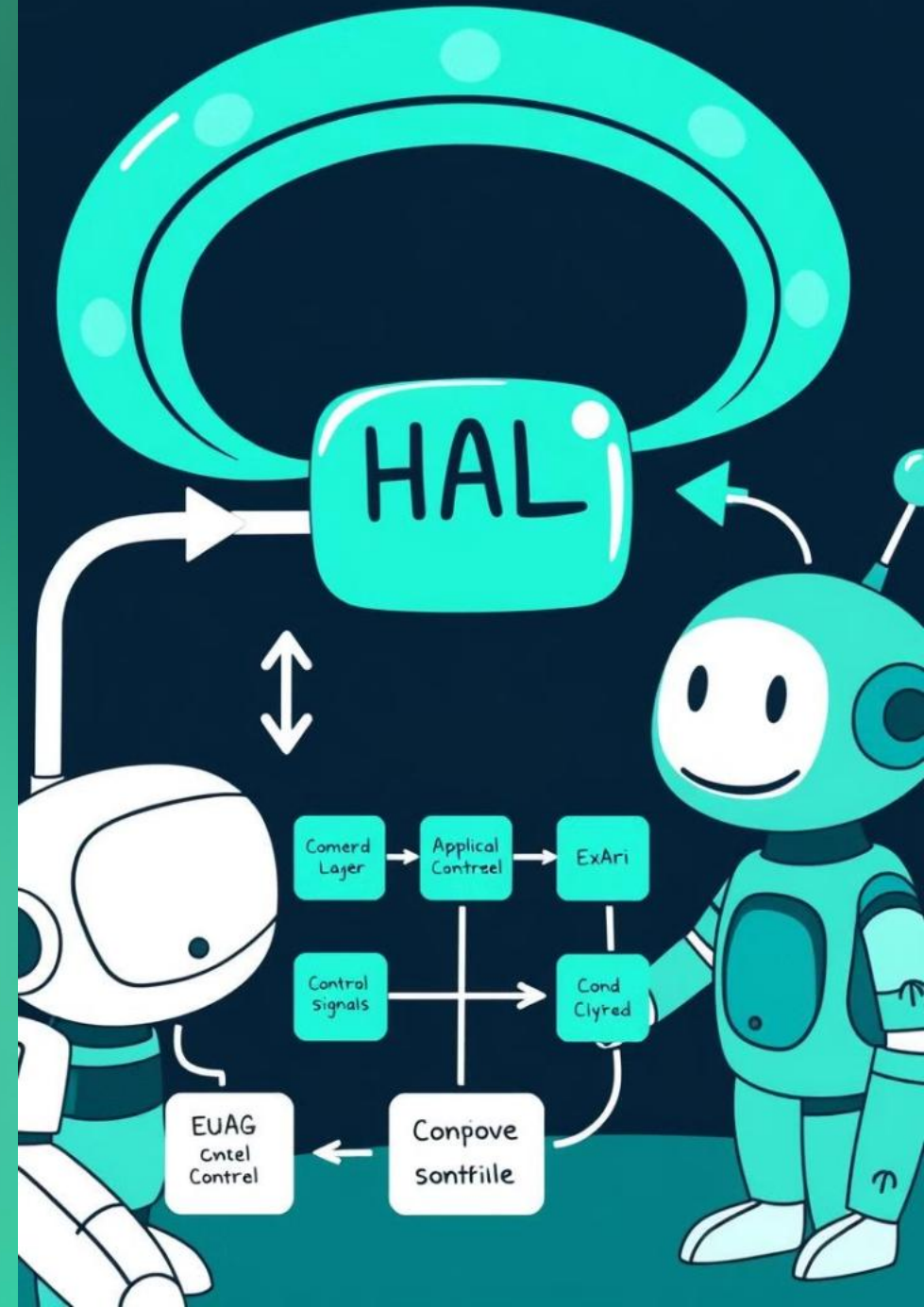
Joystick Integration: Processes user inputs.
Motor and Sensor Control: Facilitates movement and environment sensing.

2

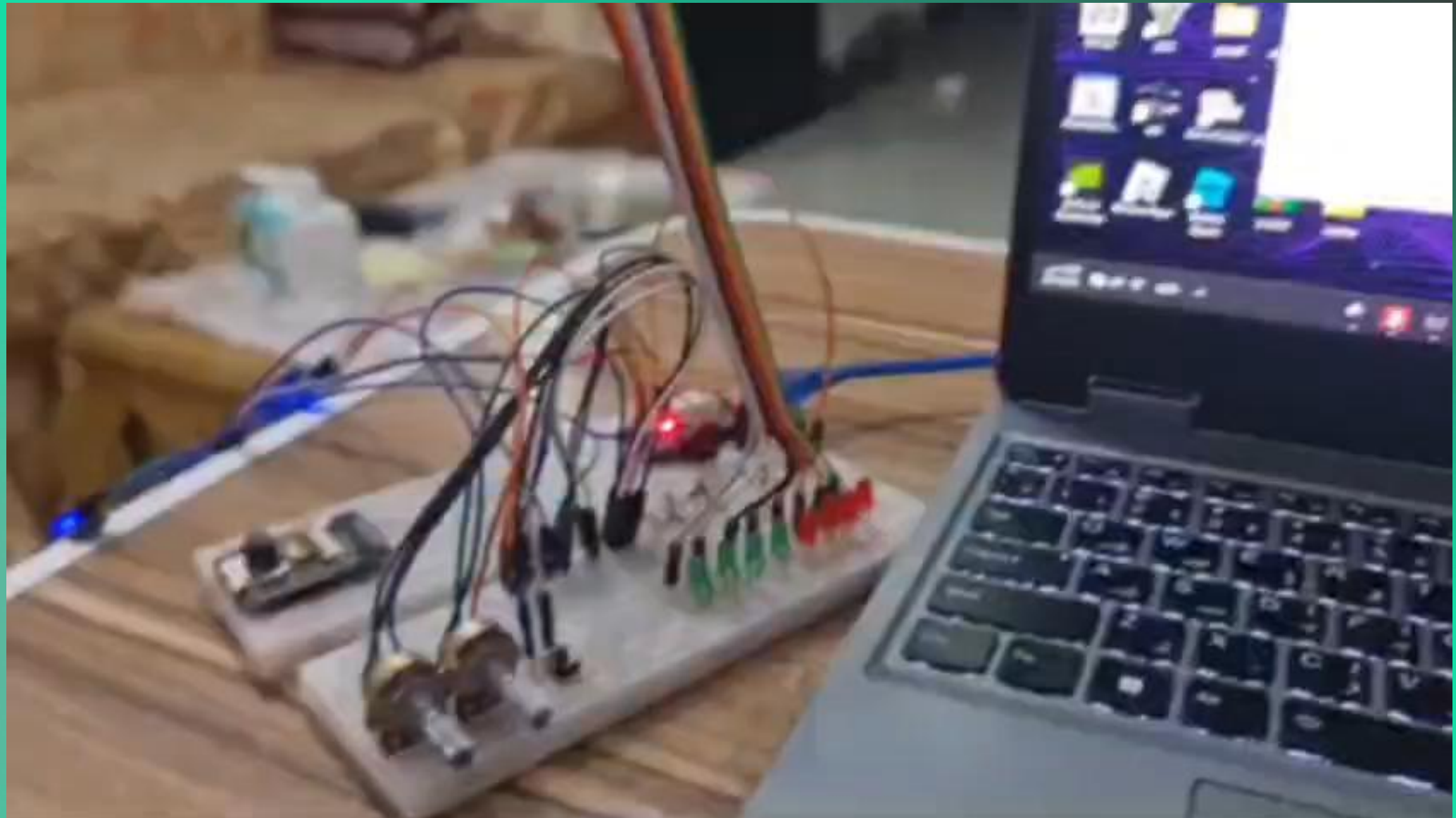
Application Layer

Coordinates system behavior, ensuring smooth operation.
Acts as the central logic controller between HAL and MCAL layers.

Summary: The design structure ensures modularity, reliability, and efficient system operation.



Sensors and motors test:





Future Trends

Special thanks

To our Volunteers



Khaled shehata



Mariam Helal



Ahmed Ibrahim



Kareem Abdelaziz



Mohamed Mansy



Hussain Rizk



Eyad Medhat

Team



Mohamed Shawky



Ahmad Yunus



Faris Elasaly

Members



Abd Al-Rahman Mahmoud



Mohamed Ibrahim



Mohamed Ramadan

